

CSE 434

SOCKET PROJECT

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Group 14

Octavio Terrazas, oateraz

Shashwat Raj, sraj28

Professor Sandeep Gupta

DEPLOYMENT AND PORTS

The demonstration uses three peer processes across two CloudLab end hosts. Node0 (10.10.1.1) runs the manager and PeerA, the leader. Node1 (10.10.1.2) runs PeerB and PeerC. Group 14 is assigned ports 8000-8499. This run uses manager port 8000; PeerA ports 8001/8002; PeerB ports 8003/8004; and PeerC ports 8005/8006.

PROGRAM

The implementation uses Python 3 standard-library modules: socket, json, threading, queue, and csv. The manager is a long-running UDP service. Each peer sends manager requests over UDP and binds a peer-to-peer UDP socket to its assigned peer port. Messages are compact UTF-8 JSON. Requests contain a type, UUID request_id, sender, and payload. Manager responses echo the request_id and include a status so the peer can match a reply to its request.

SETUP

A peer registers its name, IP address, manager port, and peer port. The manager checks that the name is alphabetic, no longer than 15 characters, and unique; it also checks that the numeric ports are in range, distinct for that peer, and not conflicting with another peer on the same IP. On success, the manager records the peer as Free and responds with SUCCESS. The peer then binds its peer-to-peer socket and starts its listener.

PeerA sends SETUP_DHT with size 3 and year 1950. The manager requires a registered leader, an idle manager, a valid size, and two other Free peers. It selects two Free peers, marks PeerA as Leader and the other members InDHT, and returns an ordered member list with the leader first. PeerA takes ring ID 0 and sends SET_ID messages assigning IDs to the other peers. Each peer records its right neighbor as the next member modulo the ring size, forming a three-peer ring.

HASHING AND RECORD DISTRIBUTION

The CSV loader skips the header and checks that each nonempty record has 14 fields. The input contains 223 storm records. For record count L , the table size is the first prime greater than $2L$. Here $L = 223$, so the first prime greater than 446 is 449. For each record, $\text{position} = \text{event_id} \bmod 449$ and $\text{target_peer_id} = \text{position} \bmod 3$.

The leader stores records assigned to ID 0 locally. For records assigned to another ID, it sends a STORE message to its immediate right neighbor, not directly to the target. A non-target peer increments the hop count and forwards the message to its own right neighbor. The target stores

the record and sends STORE_ACK to the leader. The leader transfers these records one at a time and waits for the corresponding acknowledgment.

WORKING

After distribution, the leader starts a COUNT message around the ring. Each peer adds its local record count and forwards the message to its right neighbor. When the count message returns, the leader verifies that the sum is 223 and sends DHT_COMPLETE to the manager. The manager accepts completion only from the current leader while a DHT is being built, changes its phase to Ready, logs the count map, and returns SUCCESS.

A verified CloudLab run showed three successful registrations, SETUP SUCCESS, ring IDs 0-2, 223 records, table size 449, counts of 72, 75, and 76 by ring ID, and DHT_COMPLETE SUCCESS. The counts sum to 223. The successful test log and the uploaded video are separate evidence. The two standard-library unit test checks passed on each of the fresh CloudLab nodes. Git history contains single-author, one-line commits created after source development.

DATASET

The included CSV is a 14-field projection of NOAA/NWS's public 1950 Storm Events details archive, StormEvents_details-ftp_v1.0_d1950_c20260323.[csv.gz](#). The selected fields are event ID, state, year, month, event type, county-zone type, county-zone name, direct and indirect injuries, direct and indirect deaths, property damage, crop damage, and tornado scale.

VIDEO

YouTube: <https://youtu.be/tuRaDyfUN88>

Video sections:

Introduction: 00:00-00:45

Step 3(a), compile check: 00:45-01:57

Step 3(b), show two hosts: 01:57-02:32

Step 3(c), start manager and three peers: 02:33-05:25

Step 3(d), DHT setup, counts, completion, and closing: 05:26-08:32

COMMIT HISTORY

Commits on Sep 27, 2026		
Finalize September milestone	53f1f64	
darthvader58 committed 2 minutes ago		
Add milestone design document	e9ffc1c	
darthvader58 committed 3 hours ago		
Add milestone demo checklist	fc6a70c	
darthvader58 committed 3 hours ago		
Run tests with standard library	fadb3c	
darthvader58 committed 3 hours ago		
Record CloudLab milestone results	ed69fd6	
darthvader58 committed 3 hours ago		
Document NOAA dataset provenance	64dee1	
darthvader58 committed 3 hours ago		
Add milestone dataset and tests	2487125	
darthvader58 committed 5 hours ago		
Implement peer ring distribution	1c85114	
darthvader58 committed 5 hours ago		
Implement manager registration	fc99683	
darthvader58 committed 5 hours ago		
Add storm record hashing	6ad20f9	
darthvader58 committed 5 hours ago		
Add UDP message protocol	f458e1d	
darthvader58 committed 5 hours ago		
Initialize milestone project	521646c	
darthvader58 committed 5 hours ago		